



The Solitaire Clobber game and the correducibility of k -connected graphs

Tatsuya Fujimori^a, Shun-ichi Maezawa^{b,*}, Yoshio Okamoto^c

^a Isorot Inc, 13F Shinjuku maynds tower 2-1 Yoyogi, Shibuya-ku, Tokyo, Japan

^b Department of Information Science, College of Humanities and Sciences, Nihon University, 3-25-40 Sakurajosui, Setagaya-ku, Tokyo, Japan

^c Department of Computer and Network Engineering, Graduate School of Informatics and Engineering, The University of Electro-Communications, 1-5-1 Chofugaoka, Chofu, Tokyo, Japan

ARTICLE INFO

Article history:

Received 31 March 2023

Received in revised form 30 August 2024

Accepted 3 January 2025

Available online 17 January 2025

Keywords:

Combinatorial game

Connectivity

Correducibility

ABSTRACT

Clobber is a two-player combinatorial game played on a graph with black and white stones. The Solitaire Clobber game is a single-player variant of Clobber that was introduced by Demaine et al. (2004). In the initial position, a black or white stone is placed at each vertex of an undirected graph. A player picks up a stone and clobbers a stone of different color located at an adjacent vertex; the clobbered stone is replaced by the stone that has been moved. Dantas et al. (2020) defined the k -correducibility of a graph as follows: a graph G is k -correducible if for every non-monochromatic initial configuration of stones on G and every subset S of $V(G)$ with $|S| \leq k$, there exists a procedure for moving stones such that S is empty of stones. Dantas et al. (2020) showed that a graph is ℓ -correducible if and only if it is ℓ -connected for $\ell = 1, 2$ and every $(k + \lceil \frac{k}{2} \rceil)$ -connected graph is k -correducible for $k \geq 1$.

We improve their result and show that every k -connected graph is k -correducible for $k \geq 1$. Moreover, our connectivity condition is sharp, that is, there are infinitely many $(k - 1)$ -connected graphs that are not k -correducible.

© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

1. Introduction

Clobber is a two-player combinatorial game played on a graph with black and white stones. The Solitaire Clobber game is a single-player variant of Clobber that was introduced by Demaine et al. [2]. In the initial position, a black or white stone is placed at each vertex of an undirected graph. A player picks up a stone and clobbers a stone of different color located at an adjacent vertex; the clobbered stone is replaced by the stone that has been moved. The goal of the player is to clobber as many stones as possible. Demaine et al. [2] studied some conditions for an initial configuration to allow a procedure that clobbers all stones except for one (such an initial configuration is called 1-reduced). Moreover, they showed that the problem of determining whether a given graph G and an arbitrary initial configuration is 1-reduced is NP-complete.

Dantas et al. [1] introduced a graph parameter “correducibility” and studied the relationship between the correducibility and the connectivity. The main purpose of this paper is to investigate their relationship more closely, and to improve results of Dantas et al. [1]. These results are shown in Section 2.

* Corresponding author.

E-mail addresses: maezawa.mw@gmail.com (S.-i. Maezawa), okamotoy@uec.ac.jp (Y. Okamoto).

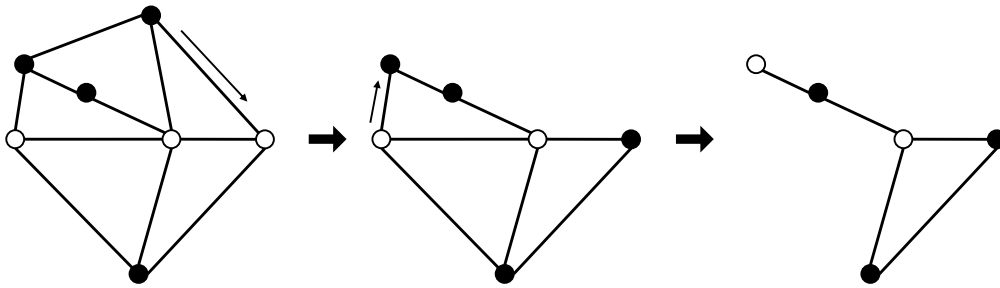


Fig. 1. The black points and the white points represent vertices in $\Phi^{-1}(0)$ and $\Phi^{-1}(1)$, respectively.

In this paper, we only consider finite undirected simple graphs. Let G be a graph. For a vertex $x \in V(G)$, we denote the degree of x in G by $\deg_G(x)$ and the set of vertices adjacent to x in G by $N_G(x)$. For a set $X \subseteq V(G)$, let $N_G(X) := \bigcup_{x \in X} N_G(x)$. A graph G is k -connected if $|G| > k$ and $G - X$ is connected for every $X \subseteq V(G)$ with $|X| < k$.

1.1. The Solitaire Clobber game and the correducibility

We define a *configuration* on a graph G to be a partial function from $V(G)$ to $\{0, 1\}$. The domain of a configuration Φ is denoted by $\text{Dom}(\Phi)$. If $\text{Dom}(\Phi) = V(G)$, then we say that Φ is an *initial configuration* on G .

A *Solitaire Clobber game* on a graph G is a finite sequence $\Phi = (\Phi_1, \Phi_2, \dots, \Phi_n)$ of configurations on G satisfying the following conditions (an example is shown in Fig. 1):

1. $V(G) = \text{Dom}(\Phi_1) \supseteq \text{Dom}(\Phi_2) \supseteq \dots \supseteq \text{Dom}(\Phi_n)$;
2. For every $1 \leq i \leq n - 1$, there exist vertices $v \in \text{Dom}(\Phi_i)$ and $w \in \text{Dom}(\Phi_{i+1})$ such that
 - (a) $vw \in E(G)$,
 - (b) $\text{Dom}(\Phi_i) = \{v\} \cup \text{Dom}(\Phi_{i+1})$,
 - (c) $\Phi_i(v) \neq \Phi_i(w)$,
 - (d) $\Phi_{i+1}(w) = \Phi_i(v)$ and $\Phi_{i+1}(u) = \Phi_i(u)$ for every $u \in \text{Dom}(\Phi_{i+1}) \setminus \{w\}$.

Intuitively, the vertices in $\text{Dom}(\Phi)$ hold stones. They are *white* if the vertices belong to $\Phi^{-1}(0)$, and *black* if they belong to $\Phi^{-1}(1)$. For vertices v and w in the second condition above, we say that a stone at v is *moved* to w . A Solitaire Clobber game is seen as a procedure of moving stones one by one in such a way that a stone can be moved only when an adjacent vertex has a stone of different color. After the movement, the stone at the adjacent vertex will be removed. Since a stone cannot be moved to a vertex without a stone, we can safely remove a vertex itself too if it does not have a stone. A natural goal of the game is to remove as many stones as possible.

Let G be a graph, $S \subseteq V(G)$, and $\Phi = (\Phi_1, \dots, \Phi_n)$ a Solitaire Clobber game on G . We say that Φ *empties* S if $\text{Dom}(\Phi_n) \cap S = \emptyset$. For a given initial configuration Φ on G , $\mathcal{S}(G, \Phi)$ denotes the collection of all subsets of $V(G)$ that are emptied by some Solitaire Clobber game on G with initial configuration Φ . If $\mathcal{S}(G, \Phi)$ contains all subsets S of $V(G)$ with $|S| \leq k$, then we say that the configuration Φ is k -correducible. The *correducibility* of an initial configuration Φ on G is defined by

$$cr(G, \Phi) := \max\{k \in \mathbb{N} \mid \Phi \text{ is } k\text{-correducible}\}.$$

If $\Phi: V(G) \rightarrow \{0, 1\}$ is a constant function, i.e., $\Phi(v) = 0$ for all $v \in V(G)$ or $\Phi(v) = 1$ for all $v \in V(G)$, then we call the configuration Φ *monochromatic*. Observe that $\mathcal{S}(G, \Phi) = \emptyset$ and $cr(G, \Phi) = 0$ for every monochromatic configuration Φ . We say that a graph G is k -correducible if every non-monochromatic initial configuration Φ on G is k -correducible. The *correducibility* of a graph G is defined by

$$cr(G) := \max\{k \in \mathbb{N} \mid G \text{ is } k\text{-correducible}\}.$$

Dantas, Marinho, and Tanushevski [1] studied the connection between the correducibility and the connectivity of graphs, and they showed the following results.

Theorem 1 (Dantas, Marinho, and Tanushevski [1]). *Let G be a graph with $|G| \geq 4$. Then, G is 2-connected if and only if G is 2-correducible.*

Theorem 2 (Dantas, Marinho, and Tanushevski [1]). Let $k \geq 1$ be an integer and let G be a k -connected graph with $|G| \geq k+2$. If Φ is an initial configuration on G satisfying $|\Phi^{-1}(i)| = 1$ for some $i = 0, 1$, then $\text{cr}(G, \Phi) \geq k$.

Theorem 3 (Dantas, Marinho, and Tanushevski [1]). Let $k \geq 1$ be an integer and let G be a $(k + \lfloor \frac{k}{2} \rfloor)$ -connected graph with $|G| \geq k+2$. Then, G is k -correducible.

Theorem 4 (Dantas, Marinho, and Tanushevski [1]). Let $k \geq 1$, let G be a $(k+m)$ -connected graph with $m \leq \lfloor \frac{k}{2} \rfloor$ and $|G| \geq k+2$, and let Φ be an initial non-monochromatic configuration on G . If $S \subseteq V(G)$ satisfies $|S| \leq k$ and $|\Phi^{-1}(i) \cap S| \leq m$ for some $i \in \{0, 1\}$, then $S \in \mathcal{S}(G, \Phi)$.

It was not known whether the connectivity condition of Theorem 3 was sharp or not. The main purpose of this paper is to study the connection between the correducibility and the connectivity of graphs more closely.

2. Main result

We obtain the following result that is stronger than Theorem 3.

Theorem 5. Let $k \geq 1$ be an integer and let G be a k -connected graph with $|G| \geq k+2$. Then, G is k -correducible.

Moreover, the connectivity condition in Theorem 5 is sharp, i.e., there are infinitely many $(k-1)$ -connected but not k -connected graphs that are not k -correducible. We will show the sharpness of Theorem 5 in Section 2.2.

2.1. Proof of Theorem 5

Before we prove Theorem 5, we briefly explain our idea to prove Theorem 5. Let S be an arbitrary subset of S of size k and let Φ be an arbitrary non-monochromatic initial configuration on G . We show $S \in \mathcal{S}(G, \Phi)$. Our aim is to find k mutually disjoint paths $P_1, P_2, \dots, P_k$ such that for $1 \leq i \leq k$,

- one of the end-vertices of P_i , say v_i , is contained in S and the other one, say w_i , is not,
- the end-vertices of P_i have different color, and
- all vertices in P_i have the same color as the color of v_i except for w_i .

If we find such paths, then we can eliminate all vertices in S by performing the following procedure for each $1 \leq i \leq k$. We move the stone at w_i to the neighborhood of v_i along P_i and move the stone at v_i to the neighborhood of v_i .

The following classical result is used in the proof of Theorem 5.

Theorem 6 (Menger [3]). Let G be a k -connected graph. Then the following hold.

- For any two vertices u and v in G , there are k mutually internally disjoint paths connecting u and v and
- For a vertex u and a vertex set Y of $V(G)$ with $|Y| = k$, there are k mutually internally disjoint paths connecting u and Y such that the end-vertices of the paths contained in Y are distinct.

Proof of Theorem 5. We prove Theorem 5 by the induction on k . For $k = 1$, Theorem 5 holds from Theorem 3 for $k = 1$. Hence we may assume that $k \geq 2$. Let S be an arbitrary subset of $V(G)$ with $|S| = k$ and let Φ be an arbitrary non-monochromatic initial configuration on G . We show $S \in \mathcal{S}(G, \Phi)$. Suppose $S \notin \mathcal{S}(G, \Phi)$. By Theorem 4 for $m = 0$, $|\Phi^{-1}(i) \cap S| \geq 1$ for all $i = 0, 1$.

Claim 2.1. There is no vertex x in $\Phi^{-1}(i) \cap S$ such that x is adjacent to a vertex in $\Phi^{-1}(1-i)$ for any $i = 0, 1$.

Proof. Suppose that there is a vertex x in $\Phi^{-1}(i) \cap S$ such that x is adjacent to a vertex y in $\Phi^{-1}(1-i)$ for some $i = 0, 1$. By symmetry, we may assume $x \in \Phi^{-1}(0)$. Let Φ_2 be a configuration on G obtained by moving the stone at x to y . Let G_2 be the graph induced by $\text{Dom}(\Phi_2)$. Then, G_2 is $(k-1)$ -connected and $|G_2| \geq k+1$. Moreover, Φ_2 is an initial configuration on G_2 . Suppose that Φ_2 is non-monochromatic. By induction hypothesis, there exists a Solitaire Clobber game $(\Phi_2, \dots, \Phi_m)$ on G_2 that empties $S \setminus \{x\}$. Then, $(\Phi, \Phi_2, \dots, \Phi_m)$ is a Solitaire Clobber game on G that empties S , a contradiction.

Thus, we may assume that Φ_2 is monochromatic. Then, $\Phi^{-1}(1) = \{y\}$ and $y \in S$ since $|\Phi^{-1}(i) \cap S| \geq 1$ for all $i = 0, 1$. Since $|G| \geq 3$, it follows $\Phi^{-1}(0) \setminus \{x\} \neq \emptyset$. We can deduce a contradiction by the above argument replacing x with y . ■

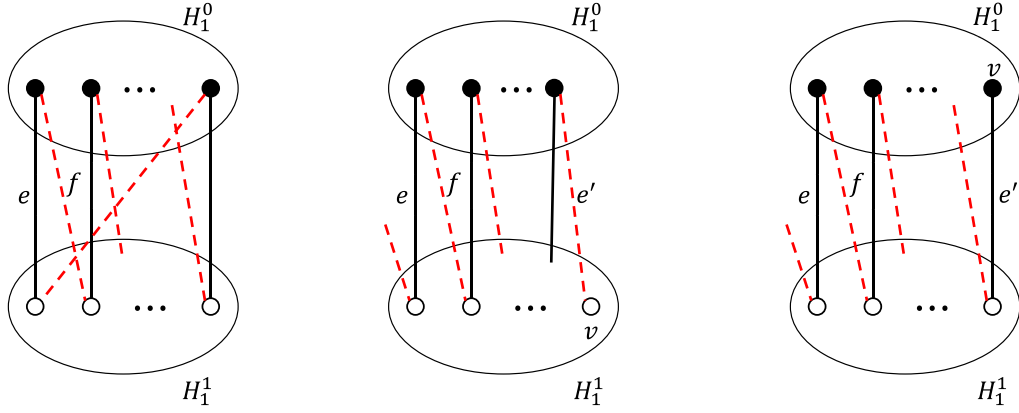


Fig. 2. The black solid lines and red broken lines represent edges in E_1^0 and E_1^1 , respectively. The left figure represents the case where P is a cycle, the middle one represents the case where P is a path and $e' \in E_1^1$, and the right one represents the case where P is a path and $e' \in E_1^0$.

Let E_{01} be the set of all edges connecting two vertices x and y such that $x \in \Phi^{-1}(0)$ and $y \in \Phi^{-1}(1)$. For $i = 0, 1$, let $H_1^i, H_2^i, \dots, H_{m_i}^i$ be the components of $G - E_{01}$ such that $V(H_j^i) \subseteq \Phi^{-1}(i)$ and H_j^i contains a vertex in S for $1 \leq j \leq m_i$. Here, m_i is the number of components of $G - E_{01}$ consisting of vertices in $\Phi^{-1}(i)$. Let H^i be the union of components of $G - E_{01}$ having no vertex in $\Phi^{-1}(1 - i)$. Since H_j^i contains a vertex in S , it follows from Claim 2.1, for $i = 0, 1$ and $1 \leq j \leq m_i$, the subgraph H_j^i contains a vertex not adjacent to a vertex in $V(H^{1-i})$ and this together with Theorem 6(i) and the k -connectivity of G shows that $|V(H_j^i)| \geq k$. Since G is k -connected, it follows from Theorem 6 (ii) that there are k internally disjoint paths connecting $u \in V(H_j^i) \cap S$ and $Y \subseteq V(H^{1-i})$ with $|Y| = k$ such that the end-vertices of the paths contained in Y are distinct. This implies that there are k independent edges between $V(H_j^i)$ and $V(H^{1-i})$ for each $i = 0, 1$ and $1 \leq j \leq m_i$. For each $i = 0, 1$ and $1 \leq j \leq m_i$ we choose an edge set E_j^i so that

- (i) E_j^i is a set of k independent edges between $V(H_j^i)$ and $V(H^{1-i})$ and
- (ii) the following value is as large as possible subject to (i):

$$\left| \bigcup_{1 \leq j \leq m_0} E_j^0 \cap \bigcup_{1 \leq j \leq m_1} E_j^1 \right|.$$

Let $E^i = E_1^i \cup E_2^i \cup \dots \cup E_{m_i}^i$ for $i = 0, 1$.

Claim 2.2. For any $i = 0, 1$, there is no edge $e \in E^i$ such that both end-vertices of e are incident with different edges in E_j^{1-i} for some $1 \leq j \leq m_{1-i}$.

Proof. Suppose that there is an edge $e \in E^i$ such that both end-vertices of e are incident with different edges in E_j^{1-i} for some $1 \leq j \leq m_{1-i}$. Without loss of generality, we may assume that $i = 0, j = 1$, and $e \in E_1^0$. Let f be an edge in E_1^1 such that f is incident with the end-vertex of e in $\Phi^{-1}(0)$. Let P be the longest path or cycle satisfying the following conditions (see Fig. 2):

- (I) One of the end-vertices of P is the end-vertex of e in $\Phi^{-1}(1)$,
- (II) P contains e and f , and
- (III) the edges in $E(P)$ appear alternately in E_1^0 and in E_1^1 .

Suppose that P is a cycle. Then, let $F_1^0 := E_1^0 \Delta E(P)$, where for two sets X and Y , $X \Delta Y := (X \setminus Y) \cup (Y \setminus X)$, which is called the symmetric difference of X and Y . Then, F_1^0 is a set of k independent edges between $V(H_1^0)$ and $V(H^1)$ and so F_1^0 satisfies condition (i) of the choice of E_1^j . However, $|F_1^0 \cap E_1^1| = |E_1^0 \cap E_1^1| + |E(P) \cap E_1^1| > |E_1^0 \cap E_1^1|$ and so

$$\left| \left(\bigcup_{2 \leq i \leq m_0} E_i^0 \cup F_1^0 \right) \cap \bigcup_{1 \leq i \leq m_1} E_i^1 \right| > \left| \bigcup_{1 \leq i \leq m_0} E_i^0 \cap \bigcup_{1 \leq i \leq m_1} E_i^1 \right|,$$

which contradicts condition (ii) of the choice of E_i^j .

Hence, we may assume that P is a path. Let v be the end-vertex of P not incident with e and let e' be the edge in $E(P)$ incident with v . We divide the proof into two cases.

Case 1. e' is in E_1^1 .

Since P is a longest path, there is no edge in E_1^0 incident with v . Let $F_1^0 := E_1^0 \triangle E(P)$. Then, F_1^0 is a set of k independent edges between $V(H_1^0)$ and $V(H^1)$ and so F_1^0 satisfies condition (i) of the choice of E_j^i . However, $|F_1^0 \cap E_1^1| = |E_1^0 \cap E_1^1| + |E(P) \cap E_1^1| > |E_1^0 \cap E_1^1|$ and so

$$\left| \left(\bigcup_{2 \leq i \leq m_0} E_i^0 \cup F_1^0 \right) \cap \bigcup_{1 \leq i \leq m_1} E_i^1 \right| > \left| \bigcup_{1 \leq i \leq m_0} E_i^0 \cap \bigcup_{1 \leq i \leq m_1} E_i^1 \right|.$$

This contradicts condition (ii) of the choice of E_1^0 .

Case 2. e' is in E_1^0 .

Since P is a longest path, there is no edge in E_1^1 incident with v . Let $F_1^1 := E_1^1 \triangle (E(P) \setminus \{e\})$. Then, F_1^1 is a set of k independent edges between $V(H_1^1)$ and $V(H^0)$ and so F_1^1 satisfies condition (i) of the choice of E_j^i . However, $|F_1^1 \cap E_1^0| = |E_1^1 \cap E_1^0| + |E(P) \cap E_1^0| > |E_1^1 \cap E_1^0|$ and so

$$\left| \bigcup_{1 \leq i \leq m_0} E_i^0 \cup F_1^1 \cap \left(\bigcup_{2 \leq i \leq m_1} E_i^1 \cup F_1^1 \right) \right| > \left| \bigcup_{1 \leq i \leq m_0} E_i^0 \cap \bigcup_{1 \leq i \leq m_1} E_i^1 \right|.$$

This contradicts condition (ii) of the choice of E_1^1 . This completes the proof of [Claim 2.2](#). ■

Recall that for $i = 0, 1$ and $1 \leq j \leq m_i$, $V(H_j^i)$ contains a vertex in S . For $i = 0, 1$ and $1 \leq j \leq m_i$, let $S_j^i := S \cap V(H_j^i)$.

Claim 2.3. *There exist k mutually disjoint paths $P_1, P_2, \dots, P_k$ such that P_i connects some vertex v_i in S and some vertex w_i with $\Phi(w) \neq \Phi(v)$ and $w \notin S$.*

Proof. We show the following statement by induction on the lexicographic order of (i, j) with $i = 0, 1$ and $1 \leq j \leq m'_i$. For (i, j) and (i', j') , if (i', j') is strictly smaller than (i, j) in lexicographic order, we write $(i', j') < (i, j)$ and if (i', j') is smaller than (i, j) or equal to (i, j) in the lexicographic order, we write $(i', j') \leq (i, j)$. Let G' be the graph induced by the edges in $\bigcup_{0 \leq i \leq 1, 1 \leq j \leq m_i} E_j^i$. We consider the following algorithm.

Step 0: Set $i := 0$ and $j := 1$.

Step 1: Take a matching M_j^i from G' that consists of $|S_j^i|$ edges in $E_j^i \cap E(G')$.

Step 2: If $(i, j) \neq (1, m_1)$, then increment (i, j) , set $G' := G' - V(M_j^i)$, and go back to Step 1. Otherwise, done.

We show the correctness of the algorithm above. To show the correctness of the algorithm, we only have to show the correctness of Step 1 in the algorithm. Since $|E_1^0| \geq |S_1^0|$, Step 1 in the algorithm is executable for $(i, j) = (0, 1)$. Let (i, j) be a pair with $(0, 1) < (i, j) \leq (1, m_1)$ such that the algorithm is executable for $(0, 1) \leq (i', j') < (i, j)$. Let $G_j^{i'} := G' - \bigcup_{(0, 1) \leq (i'', j'') < (i', j')} V(M_{j''}^{i''})$ for $(0, 1) < (i', j') \leq (i, j)$. Since $E(M_{j'}^{i'}) \subseteq E_j^{i'}$ for each $(0, 1) \leq (i', j') < (i, j)$, it follows from [Claim 2.2](#) that each edge in $M_{j'}^{i'}$ is incident with at most one edge in E_j^i and so the number of deleted edges in $E_j^i \cap E(G_j^{i'})$ by deleting both end-vertices of an edge in $M_{j'}^{i'}$ in Step 2 is at most one. Hence we obtain the number of deleted edges in Step 2 is at most $|S_j^{i'}|$, that is

$$|E_j^i \cap E(G_j^{i'} - V(M_{j'}^{i'}))| \geq |E_j^i \cap E(G_j^{i'})| - |S_j^{i'}|.$$

Since $|S| = \sum_{(0, 1) \leq (i, j) \leq (1, m_1)} |E_j^i| = k$, we obtain

$$|E_j^i \cap E(G_j^{i'})| \geq |E_j^i| - \sum_{(0, 1) \leq (i', j') < (i, j)} |S_j^{i'}| = k - \sum_{(0, 1) \leq (i', j') < (i, j)} |S_j^{i'}| \geq |S_j^i|.$$

Hence, we can take a matching from $G_j^{i'}$ that consists of $|S_j^i|$ edges in $E_j^i \cap E(G_j^{i'})$ and so Step 1 in the algorithm is executable for $(0, 1) \leq (i, j) \leq (1, m_1)$.

Since G is k -connected, for $(0, 1) \leq (i, j) \leq (1, m_1)$, there are $|S_j^i|$ mutually disjoint paths connecting S_j^i and $V(M_j^i) \cap V(H_j^i)$ in H_j^i and we obtain desired paths by combining these paths and M_j^i for $(0, 1) \leq (i, j) \leq (1, m_1)$. See [Fig. 3](#). ■

For $1 \leq i \leq k$, let v_i and w_i be the end-vertices of P_i with $w_i \notin S$ and let x_i be the neighbor of v_i in P_i . Consider the Solitaire Clobber game that moves the stone at w_i to x_i along P_i and moves the stone at v_i to x_i for $1 \leq i \leq k$. This game empties S , a contradiction. ■

2.2. Sharpness of [Theorem 5](#)

We show that there are infinitely many $(k - 1)$ -connected graphs that are not k -connected and not k -correducible.

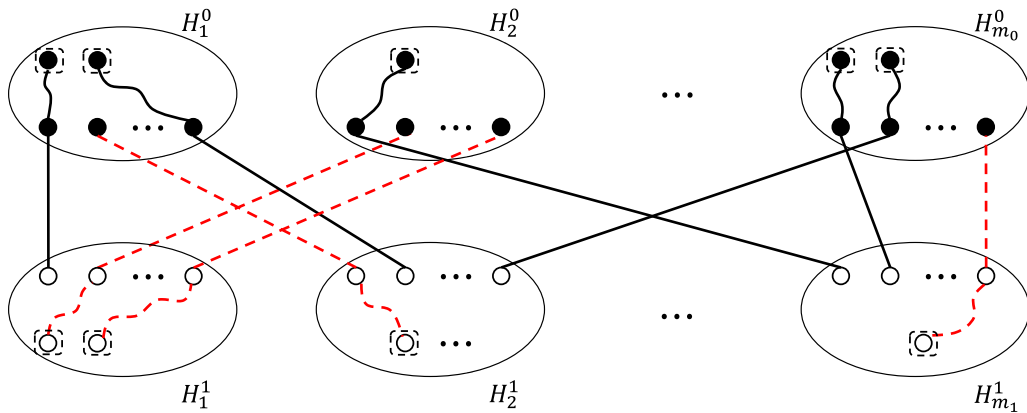


Fig. 3. The black solid lines and red broken lines represent edges in M_i^0 , $1 \leq i \leq m_0$, and M_j^1 , $1 \leq j \leq m_1$, respectively. The black wavy solid lines and red wavy broken lines represent paths connecting S_i^0 and $V(M_i^0) \cap V(H_i^0)$ in H_i^0 ($1 \leq i \leq m_0$) and connecting S_j^1 and $V(M_j^0) \cap V(H_j^1)$ in H_j^1 ($1 \leq j \leq m_1$), respectively. The points surrounded by dotted squares represent vertices in S . The black points and the white points represent vertices in $\Phi^{-1}(0)$ and $\Phi^{-1}(1)$, respectively.

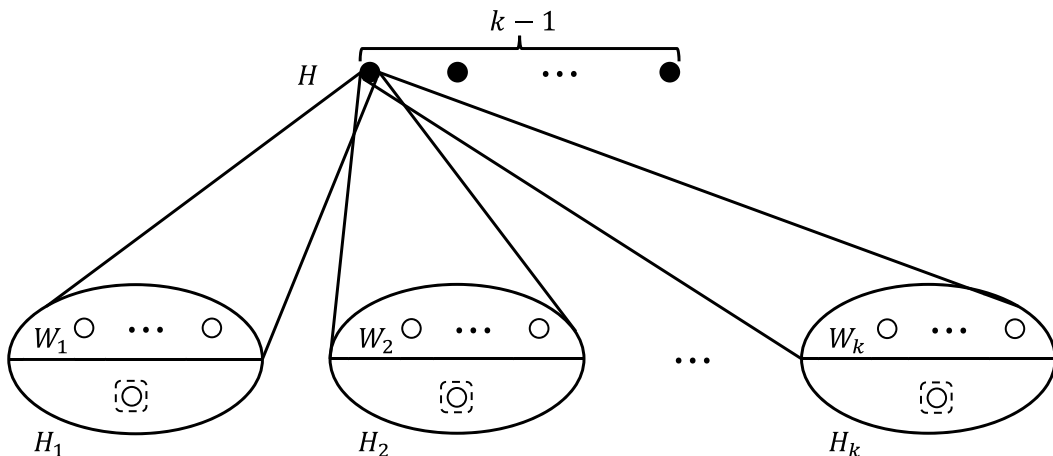


Fig. 4. The black points and white points represent vertices in $\Phi^{-1}(0)$ and $\Phi^{-1}(1)$, respectively. The points surrounded by dotted squares represent vertices in S . To avoid complication, a part of the edges connecting H and $\cup_{1 \leq i \leq k} V(H_i)$ is shown.

Let H be a graph consisting of $k - 1$ independent vertices and let H_i be a graph isomorphic to a complete graph of order at least k for $1 \leq i \leq k$. Let W_i be a subset of $V(H_i)$ with $|W_i| = k - 1$ for $1 \leq i \leq k$. Let G be the graph obtained from $H, H_1, \dots, H_k$ by adding all edges between H and $\cup_{1 \leq i \leq k} W_i$. Let $\Phi: V(G) \rightarrow \{0, 1\}$ be an initial configuration on G such that $\Phi(x) = 0$ for every $x \in V(H)$ and $\Phi(y) = 1$ for every $y \in \cup_{1 \leq i \leq k} V(H_i)$. Then, G is $(k - 1)$ -connected but not k -connected since $G - V(H)$ is disconnected. Let S be a vertex set such that $S \cap V(H) = \emptyset$ and $|S \cap (V(H_i) \setminus W_i)| = 1$ for $1 \leq i \leq k$. See Fig. 4.

Intuitively, we need to move at least one vertex in $V(H)$ to delete one vertex in S in G and if so there is no Solitaire Clobber game on G that empties S . However, we should care about any Solitaire Clobber game on G and it is tough to check each and every one of them. In order to overcome this, we introduce a potential function f mapping from configurations on G to integers, which determines whether a configuration contains a vertex in S or not. More precisely, a configuration contains a vertex in S if and only if f returns a positive integer. We will show that f returns a positive integer for any given configuration in any Solitaire Clobber game on G .

We show that there is no Solitaire Clobber game on G that empties S . Let $\Phi = (\Phi_1, \Phi_2, \dots, \Phi_m)$ be an arbitrary Solitaire Clobber game on G . For $1 \leq i \leq m$, we define

$$N_i := |V(H) \cap \text{Dom}(\Phi_i)| + |\{j \mid 1 \leq j \leq k, V(H_j) \cap \Phi_i^{-1}(0) \neq \emptyset \text{ and } V(H_j) \cap S \neq \emptyset\}|$$

$$f(i) := |S \cap \text{Dom}(\Phi_i)| - N_i.$$

We prove $f(i) \geq 1$ holds for $1 \leq i \leq m$. Since $N_i \geq 0$ for $1 \leq i \leq m$, if the inequality holds for $1 \leq i \leq m$, then there is no Solitaire Clobber game on G that empties S .

We show $f(i) \geq 1$ by the induction on $1 \leq i \leq m$. For convenience, let

$$M_i := |\{j \mid 1 \leq j \leq k, V(H_j) \cap \Phi_i^{-1}(0) \neq \emptyset \text{ and } V(H_j) \cap S \neq \emptyset\}|$$

for $1 \leq i \leq m$. Since $|S| = k$ and $N_1 = |V(H) \cap \text{Dom}(\Phi_1)| + M_1 = |V(H)| + 0 = k - 1$, it holds $f(1) \geq 1$. Hence, we may assume that $i \geq 2$ and $f(i') \geq 1$ for $1 \leq i' < i$. We divide the proof into cases by a way of the transition from Φ_{i-1} to Φ_i .

Case 1. In the transition from Φ_{i-1} to Φ_i , a stone at a vertex in $V(H)$ moves to a vertex in $V(H_j)$ for some $1 \leq j \leq k$.

In this case, $N_i = |V(H) \cap \text{Dom}(\Phi_i)| + M_i \leq |V(H) \cap \text{Dom}(\Phi_{i-1})| - 1 + M_{i-1} + 1 = N_{i-1}$. Since $N_G(V(H)) \cap S = \emptyset$, it holds $|S \cap \text{Dom}(\Phi_{i-1})| = |S \cap \text{Dom}(\Phi_i)|$. Hence, $f(i) \geq f(i-1) \geq 1$.

Case 2. In the transition from Φ_{i-1} to Φ_i , a stone at a vertex in $V(H_j)$ moves to a vertex in $V(H)$ for some $1 \leq j \leq k$.

In this case, $N_i = |V(H) \cap \text{Dom}(\Phi_i)| + M_i \leq |V(H) \cap \text{Dom}(\Phi_{i-1})| + M_{i-1} = N_{i-1}$. Since $N_G(V(H)) \cap S = \emptyset$, it holds $|S \cap \text{Dom}(\Phi_{i-1})| = |S \cap \text{Dom}(\Phi_i)|$. Hence, $f(i) \geq f(i-1) \geq 1$.

Case 3. In the transition from Φ_{i-1} to Φ_i , a stone at a vertex x in $V(H_j)$ moves to a vertex y in $V(H_j)$ for some $1 \leq j \leq k$.

Suppose $x \in S$. Then,

$$N_i = |V(H) \cap \text{Dom}(\Phi_i)| + M_i = |V(H) \cap \text{Dom}(\Phi_{i-1})| + M_{i-1} - 1 = N_{i-1} - 1$$

and $|S \cap \text{Dom}(\Phi_i)| = |S \cap \text{Dom}(\Phi_{i-1})| - 1$. Hence,

$$f(i) = |S \cap \text{Dom}(\Phi_i)| - N_i = |S \cap \text{Dom}(\Phi_{i-1})| - 1 - (N_{i-1} - 1) = f(i-1) \geq 1.$$

We may assume $x \notin S$. Suppose $x \in \Phi^{-1}(1)$. Then,

$$N_i = |V(H) \cap \text{Dom}(\Phi_i)| + M_i \leq |V(H) \cap \text{Dom}(\Phi_{i-1})| + M_{i-1} = N_{i-1}$$

and $|S \cap \text{Dom}(\Phi_{i-1})| = |S \cap \text{Dom}(\Phi_i)|$. Hence,

$$f(i) = |S \cap \text{Dom}(\Phi_{i-1})| - N_i \geq |S \cap \text{Dom}(\Phi_i)| - N_{i-1} = f(i-1) \geq 1.$$

Suppose $x \in \Phi^{-1}(0)$. Then,

$$N_i = |V(H) \cap \text{Dom}(\Phi_i)| + M_i = |V(H) \cap \text{Dom}(\Phi_{i-1})| + M_{i-1} = N_{i-1}$$

and $|S \cap \text{Dom}(\Phi_{i-1})| = |S \cap \text{Dom}(\Phi_i)|$. Hence,

$$f(i) = |S \cap \text{Dom}(\Phi_{i-1})| - N_i = |S \cap \text{Dom}(\Phi_i)| - N_{i-1} = f(i-1) \geq 1.$$

This completes the proof that $f(i) \geq 1$ holds for $1 \leq i \leq m$ and we conclude that there is no Solitaire Clobber game on G that empties S .

Acknowledgments

The second author's work (S. Maezawa) is supported by JSPS KAKENHI, Japan Grant Number JP20H05795 and JSPS KAKENHI Grant Number JP22K13956. The third author's work (Y. Okamoto) is supported by JSPS KAKENHI, Japan Grant Number JP20H05795 and JSPS KAKENHI, Japan Grant Number JP23K10982.

Data availability

No data was used for the research described in the article.

References

- [1] S. Dantas, R. Marinho, S. Tanushevski, The Solitaire Clobber game and correducibility of graphs, *Discrete Appl. Math.* 281 (2020) 144–150.
- [2] E.D. Demaine, M.L. Demaine, R. Fleischer, Solitaire Clobber, *Theoret. Comput. Sci.* 313 (2004) 325–338.
- [3] K. Menger, Zur allgemeinen Kurventheorie, *Fund. Math.* 10 (1927) 95–115.